Requirements-based testing of neural networks.

A critical review of RBT4DNN, its empirical evidence, and three exploratory additional analyses.

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From a requirement to credible evidence.

01	02	03	04	05	06
Problem statement	**Background**	**Current approach**	**Methodology**	**Current results**	**Critical review**
Why test by requirement?	Terms and assumptions	RBT4DNN overview	Data, labels, generator adaptation	RQ1–RQ6	Additional analyses + future scope

WHY REQUIREMENTS?

HOW DOES RBT4DNN WORK?

HOW STRONG IS THE EVIDENCE?

Structure follows the seminar brief; appendix begins after the conclusion.

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING

Known behavior

A specification maps an input condition to an expected output. Test oracles can check the result directly.

DNN TESTING

Learned behavior

The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.

THE MISSING LINK

Semantic scenarios

Developers need evidence for statements such as "a very thick 7 must still be classified as 7."

GLOBAL METRIC
How often is the complete test set correct?

→

SEMANTIC SLICE
Which rare inputs does the score contain?

→

REQUIREMENT VERDICT
Does the model satisfy this specific contract?

[1] §1–§2.5 · "model under test" means the evaluated neural network

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 · authors claim the first requirement-driven DNN test-input generator

Four domains test four different semantic challenges.

MNIST · 7 REQUIREMENTS

MNIST artifact samples
Source-linked in interactive deck

Digits

Measured height, thickness, and lean. A controlled starting point.

Output: digit class

CELEBA-HQ · 7 REQUIREMENTS

CelebA-HQ artifact samples
Source-linked in interactive deck

Faces

Ambiguous attributes labeled by visual question answering or humans.

Output: eyeglasses decision

SYNTHETIC DRIVING (SGSM) · 7

SGSM artifact samples
Source-linked in interactive deck

Driving

Scene relations linked to steering and acceleration constraints.

Output: numeric control

IMAGENET · 4 REQUIREMENTS

ImageNet artifact samples
Source-linked in interactive deck

Animals

Morphology linked to allowed classes in the WordNet taxonomy.

Output: class set

[1] Table 2 and §4.1.1–§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, **then** the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN is a four-stage semantic pipeline.

01 · LABEL

Pixels → terms

Attach domain facts such as very thick, eyeglasses, or same lane.

→

02 · FILTER

Terms → subset

Keep original images satisfying one requirement precondition.

→

03 · ADAPT

Subset → LoRA

Fine-tune FLUX with a small adapter for that precondition.

→

04 · TEST

Images → evidence

Generate tests, run the model under test, and check the required output.

M3 LABELS	FILTERED SUBSET	TARGETED GENERATOR	EVIDENCE
digit = 7 · thickness = very thick	only matching MNIST images	one M3 LoRA adapter	precondition match + output pass or fail

[1] Fig. 2 and §3 · detailed requirement notation moved to the appendix

Glossary facts are created first—and checked later with different models.

TRAINING-TIME SOURCE	MNIST measurements	Thickness, height, and lean are measured, then numeric ranges become terms.	“very thick”
TRAINING-TIME SOURCE	SGSM scene graphs	Entities and graph relations are traversed relative to the ego vehicle.	“same lane”
TRAINING-TIME SOURCE	MiniCPM visual Q&A	Yes/no prompts label CelebA-HQ and ImageNet attributes; CelebA also has human labels.	“eyeglasses”

EVALUATION-TIME CHECK: Generated image → **ResNet-101 glossary classifiers** One binary classifier per term; mean accuracy 94.5%. → **All required terms true?** Precondition match for RQ1; term distributions for RQ3.

[1] §3.2 and §4.1.6 · separate roles, but not independent semantic ground truth; C4 classifier accuracy 70.8%

Filtering concentrates generation on a rare semantic region.

FROZEN BACKBONE

$$W$$

FLUX.1-dev: a pretrained 12-billion-parameter image generator.

+

TRAINABLE UPDATE

$$BA$$

Small low-rank matrices adapt the model to one filtered subset.

BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region
SCALING COST	One adapter must be trained and maintained per requirement

[1] §3.3 and §4.1.4 · LoRA equation simplified; latent-space coverage remains future work

The baselines are not interchangeable.

PRIMARY METHOD **Requirement-specific LoRA** Filtered precondition subset. The central RBT4DNN configuration.	MNIST CONSISTENCY ABLATION **All-data LoRA** Same domain and prompting, but the adapter sees the complete dataset.
DISTRIBUTIONAL QUALITY **Prompt-only FLUX** No domain adaptation. Used for distributional quality comparisons.	MNIST TESTING BASELINES **DeepHyperion + transformations** Established test generators that are not conditioned on the requirement.

Hardware, sampling counts, and adapter times are reproducibility details—not the experimental claim. They are in the appendix.

[1] §4.1.4–§4.1.6 · no truly comparable requirement-targeted baseline existed

Six questions form a two-stage evidence argument.

Validate the generated inputs

RQ1–RQ3

RQ1	Consistency	Does the image satisfy the precondition?
RQ2	Plausibility	Is its distribution close to matching training images?
RQ3	Variation	Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4–RQ6

RQ4	Comparison	Are targeted outcomes more informative than baselines?
RQ5	Violations	Do tests reveal credible postcondition failures?
RQ6	Diagnosis	Do failures expose repeatable decision patterns?

[1] §4.1 · one slide per RQ keeps the evidence and its boundary together

Targeted LoRA makes substantially more on-target tests.

AUTHORS' ANSWER

92.1% mean precondition consistency across 21 requirements and three datasets.

Domain means differ: CelebA-HQ automated visual-question labels 95.8%; human labels 96.1%; SGSM is weaker at 82.3%.

Setting	Model pass	Precondition match	
M3 · targeted very thick 7	69.4%	95.8%	Most outcomes address the requested scenario.
M3 · all data same prompt	99.6%	7.4%	The excellent pass rate mostly describes ordinary sevens.

MNIST Fig. 3 also includes DeepHyperion and image transformations: neither targets the requirement, and both achieve low match.

[1] §4.2.1 and Fig. 3 · glossary-classifier mean accuracy 94.5%; it remains a proxy, not ground truth

KID asks whether two image distributions occupy similar feature space.

RESULT

All MNIST and SGSM LoRA rows beat prompt-only FLUX; average reported reduction: 77.09%.

BOUNDARY

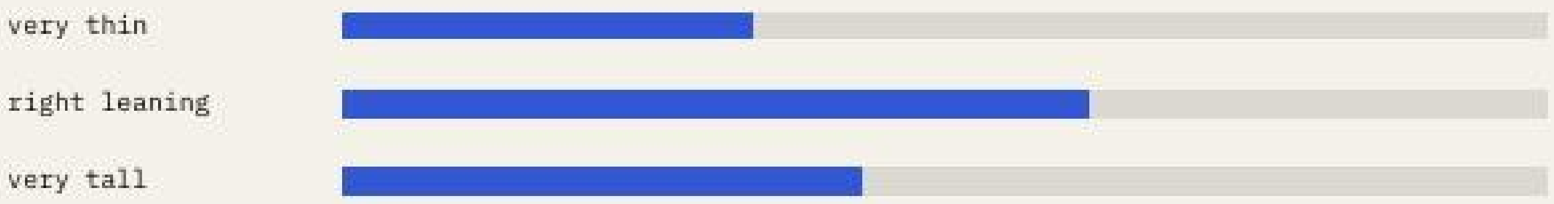
CelebA-HQ is mixed. KID is a distributional proxy, and several reference subsets contain fewer than 300 images.

[1] §4.2.2 and Table 3 · lower KID is not an image-level probability or proof of semantic validity

RQ3 preserves measured feature frequencies—not unrestricted image diversity.

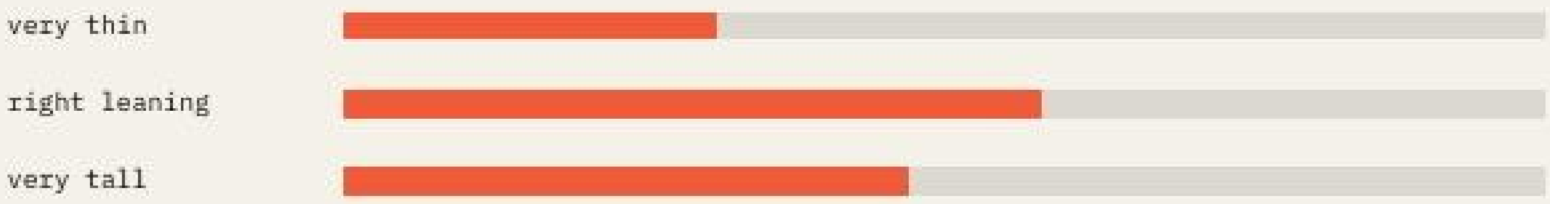
FILTERED TRAINING SUBSET

Other glossary features



GENERATED TESTS

Glossary-classifier frequencies



RESULT

MNIST and SGSM targeted-LoRA distributions were more than three times closer on average than prompt-only FLUX; CelebA-HQ improved about 32%.

NOT MEASURED

Duplicates, memorization, pixel diversity, within-term mode collapse, and systematic latent-space coverage.

[1] §4.2.3 and Table 4 · schematic bars explain the metric; they are not reported measurements

Check relevance before interpreting a pass or failure.

GATE 1 · INPUT

Does this image actually show a very thick 3?

DeepHyperion and transformations generate recognizable digits, but they are not conditioned on M2 and rarely reach its semantic precondition.

→

GATE 2 · OUTPUT

Only then does the classifier outcome address M2.

RBT4DNN trades a superficially higher pass rate for much more requirement-relevant evidence.

Narrow conclusion: requirement-aware generation is a better basis for these MNIST requirements than the two tested untargeted baselines—not universally superior DNN testing.

[1] §4.3.1, Fig. 7 and Table 5 · no truly comparable requirement-targeted baseline was available

Reported violations appeared in 19 of 21 requirements.

330

failing images independently assessed by multiple authors and aggregated.

3.3%

mean estimated false-positive rate across the reviewed requirements.

SAMPLE	11 requirements	30 failures reviewed for each requirement whose pass rate was below 90%.
QUALITY	All realistic	Worst estimated false-positive rate was 11.5%; assessor disagreement on precondition consistency was 1.7%.
INSIGHT	Tests expose more than model errors	S2 suggested an underspecified distance condition; S3 exposed generation and labeling difficulty.

[1] §4.3.2 and Fig. 7 · postcondition violations are observed; internal model faults are not localized

Failures reveal repeatable patterns—but not their causes.

1,486

ImageNet failures examined across the exploratory study.

3

high-accuracy models showed concentrated, model-specific predictions.

OBSERVED	Prediction clusters	Each classifier repeatedly favored particular incorrect labels for some requirements.
ASSOCIATED	Image patterns	Manual inspection found partial animals, background objects, and multi-object scenes.
BOUNDARY	Exploration, not causality	The patterns motivate hypotheses about ImageNet's single-label supervision; they do not prove causal mechanisms.

[1] §4.3.3, Fig. 8 and Tables 6–7 · pass rates span approximately 93.5%–99.5%

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING

Better tests

Improve generator training and adaptation to reveal more observed failures and reduce false positives.

COVERAGE EVIDENCE

Latent coverage

Adapt systematic latent-space coverage when test generation does not reveal a failure.

GENERALIZATION

More domains

Expand driving-like safety datasets and explore broader robustness and functional requirements.

ASSURANCE BOUNDARY

Automated labels, selected datasets, narrow baselines, and small KID subsets limit how far the evidence generalizes.

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

A reported model failure is evidence only when the input is valid.

PRECONDITION INVALID · MODEL PASSES **Irrelevant pass** The output says nothing about the stated requirement.	PRECONDITION INVALID · MODEL FAILS **Off-target false positive** A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES **Valid requirement pass** Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS **Credible counterexample** This joint event is the paper's strongest testing evidence.

SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Fair critique: the paper did perform joint validation. Its generalization remains limited by author assessment, fixed samples, no reported confidence intervals, and no external or blinded annotators.

The released MNIST fields closely track the paper table.

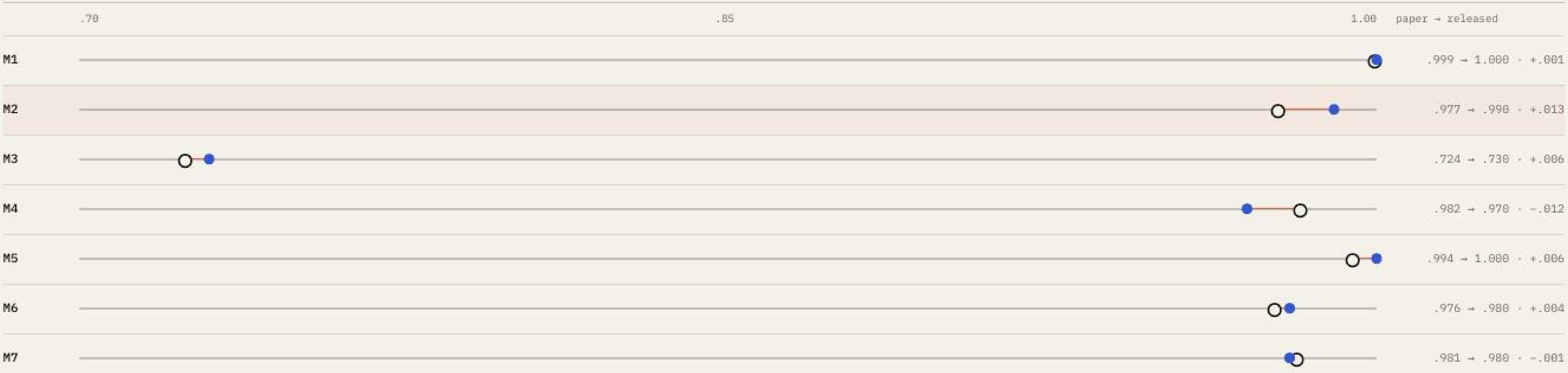
0.013

largest absolute difference across M1–M7

ARTIFACT CONSISTENCY AUDIT

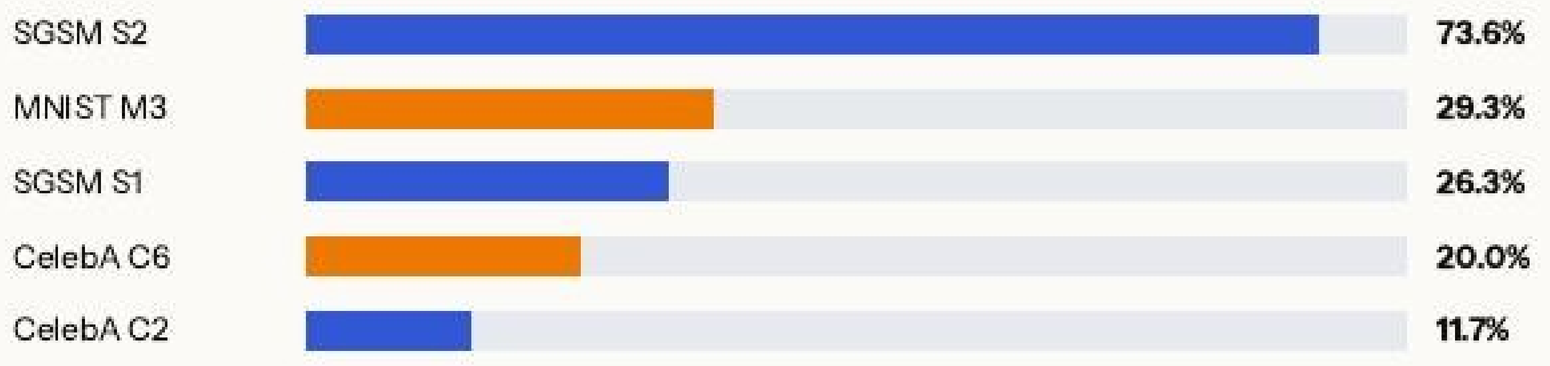
Seven stored fields were compared; inference, generation, and LoRA training were not rerun.

○ paper · ● released



Use the proxy to prioritize audits—not to estimate valid failures.

Independence-proxy ranking



Diagnostic only · unconditional P(match) × P(fail) · not an observed joint rate

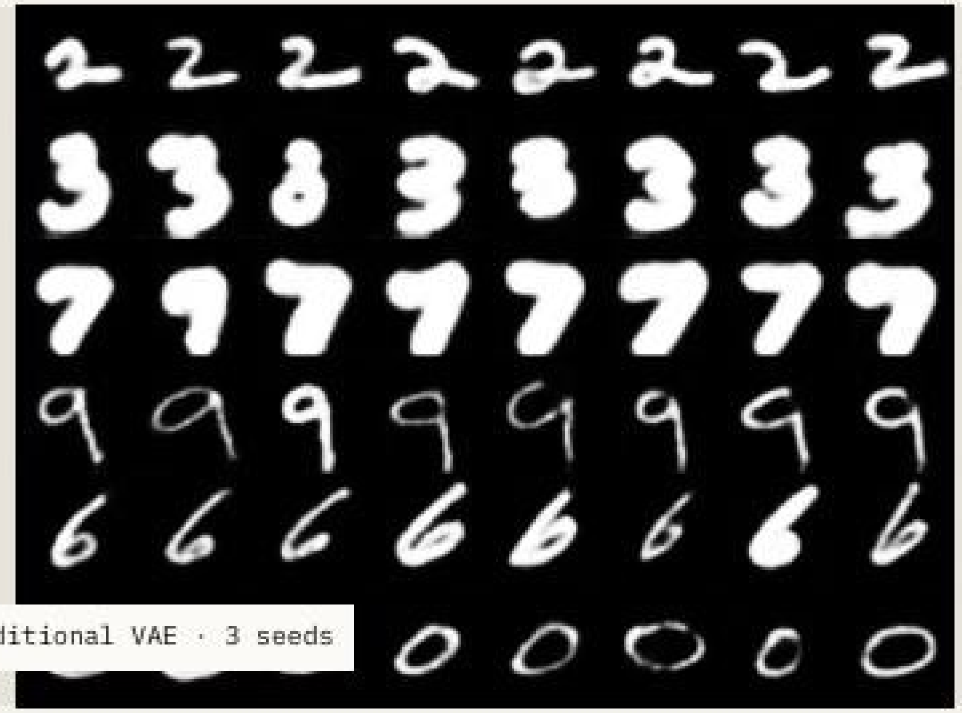
SMALL EXTERNAL SEMANTIC CHECK

.738

31 of 42 sampled images were judged precondition-valid by one multimodal model.

Use: decide where independent joint review is most valuable.

The product assumes independence. The paper's conditional human audit among failures is methodologically stronger; this LLM check is only a second opinion.



MNIST · shared conditional VAE · 3 seeds

.945

mean classifier pass rate

.942

paper v3 mean outcome rate, M1–M6

A shared generator is promising on MNIST—but not yet a general replacement.

Weak case	M3 performance	.743
CelebA	generated-image alignment	.613
Evaluator	CelebA evaluator accuracy	.636

Conclusion: the similar classifier pass rate is a promising pilot in a simple domain. Evaluation pipelines differ, and it says nothing yet about precondition validity, realism, diversity, FLUX + LoRA, or natural images.

Requirements make tests relevant. Validity makes failures credible.

WHAT THE PAPER ESTABLISHES	WHAT REMAINS OPEN
Structured requirements can guide realistic, diverse, and scenario-specific neural-network test generation across four domains.	Broader independent joint audits, shared generators, alternative adaptation methods, and end-to-end replication at larger scale.

Overall: a strong and original testing direction, with an incomplete assurance argument—and clear opportunities for the next study.

CSFR and SFFR formalize two different expectations.

CONDITIONAL SEMANTIC FEATURE ROBUSTNESS

$$\forall x \in X: \varphi(x) \Rightarrow \forall i \in I: N(x) = N(x \oplus i)$$

Inside scenario ϕ, varying a specified semantic feature within the condition must not change the output.

SEMANTIC FEATURE FUNCTIONAL REQUIREMENT

$$\forall x \in X: \varphi_X(x) \Rightarrow \varphi_Y(N(x))$$

Whenever the input precondition holds, the output must satisfy the postcondition.

X / Y	I	⊕	∀
input / output spaces	semantic variations	apply a variation	for every

[1] Definitions 2–3, §3.1

The study operationalizes 25 structured requirements.

MNIST	M1–M7 · digit class × height / thickness / lean
CELEBA-HQ	C1–C7 · eyeglasses × hair / hat / facial hair
SGSM	S1–S7 · driving-code conditions → steering / acceleration
IMAGENET	I1–I4 · animal morphology → WordNet taxon

M1–M7

MNIST

Digit identity plus height, thickness, or lean → preserve the digit label.

C1–C7

CelebA-HQ

Eyeglasses plus a second facial attribute → preserve the eyeglasses decision.

S1–S7

SGSM

Spatial driving relations → constrain steering or acceleration.

I1–I4

ImageNet

Animal morphology → require an output inside a WordNet taxon.

[1] Table 2 · requirement families redrawn for question-time legibility

Each metric supports a different link in the argument.

RQ1 · PRECONDITION MATCH

Boolean glossary agreement

All classifiers required by the precondition must return the expected terms.

RQ2 · KID ↓

Kernel Inception Distance

Distribution distance between generated images and the matching training subset.

RQ3 · JS ↓

Jensen–Shannon divergence

Distance between generated and training semantic-feature distributions.

RQ4–RQ6 · OUTCOMES

Pass / fail / false positive

Postcondition result plus manual assessment of sampled failing inputs.

[1] §4.1.6 · mean glossary-term classifier accuracy 94.5%; C4 outlier 70.8%

What the additional experiments do—and do not—establish.

Artifact	Compares released fields with the paper table	No pipeline rerun
Valid failure	Independence heuristic ranks evidence yield	Not observed joint labels
LLM audit	Second-opinion semantic review	Not ground truth
Shared generator	Postcondition pass-rate pilot with a conditional VAE	Not a substitute

Reason for explicit boundaries: scientific strength comes from matching each conclusion to the evidence actually produced.

Adapter cost scales with the requirement set.

DERIVED SERIAL TOTAL

≈24.7 A100-hours for 21 primary adapters.

MEASURED	Reported training times on one NVIDIA A100 with 80 GB
EXCLUDED	Image labeling, generation, evaluation, engineering, and ImageNet adapters
ILLUSTRATIVE ONLY	\$3/GPU-hour · \$25/engineer-hour · \$0.03/validated image
DO NOT CLAIM	Dollar-per-failure rankings as measured economic results

Use in discussion: the pricing notebook is a sensitivity scenario. Its strongest conclusion is qualitative: per-requirement adaptation creates linear training and maintenance overhead.

[1] §4.1.4 · seminar cost-analysis assumptions are illustrative, not invoices

References.

[1] **N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer.** "RBT4DNN: Requirements-based Testing of Neural Networks." arXiv:2504.02737, v3, 2025.

[2] **E. J. Hu et al.** "LoRA: Low-Rank Adaptation of Large Language Models." ICLR, 2022.

[3] **RBT4DNN artifact.** github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4] **Seminar experiments.** Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗